

PITTSBURGH INTL, PA, USA

WMO: 725200

Lat: 40.5317N

Lon: 80.2172W

Elev: 1203

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94823

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.3	(c) 9.2	(d) -4.4	(e) 4.5	(f) 7.4	(g) -0.4	(h) 5.6	(i) 11.7	(j) 25.3	(k) 29.6	(l) 23.6	(m) 26.9	(n) 7.1	(o) 260	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	88.9	71.9	86.4	70.8	84.3	69.6	75.0	84.2	73.5	81.8	72.3	80.3	8.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	124.8	79.3	70.9	119.1	77.9	69.6	113.9	76.5	39.2	84.5	37.8	82.0	36.6	80.2	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.1	17.0	-1.5	91.8	6.4	3.1	-6.0	94.1	-9.8	95.9	-13.3	97.7	-18.0	100.0
			-2.2	77.6	6.2	1.7	-6.7	78.8	-10.4	79.8	-13.9	80.8	-18.4	82.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	28.6	31.5	39.9	52.0	61.4	69.7	73.1	71.9	65.4	53.8	42.8	34.0		
	DBStd	18.04	12.30	11.36	11.23	9.59	8.11	6.14	4.91	4.89	6.84	8.61	9.36	10.10		
	HDD50	2467	671	527	355	90	8	0	0	0	0	58	254	504		
	HDD65	5518	1128	939	781	401	169	27	3	6	77	360	666	961		
	CDD50	3242	8	9	41	149	363	590	717	679	462	177	38	9		
	CDD65	816	0	0	2	11	58	168	255	220	88	14	0	0		
	CDH74	6489	0	1	13	123	546	1335	2035	1664	681	89	2	0		
	CDH80	1802	0	0	0	18	119	383	621	474	175	12	0	0		
(14) Wind	WSAvg	7.4	8.9	8.8	9.0	8.9	7.3	6.6	5.9	5.6	5.8	6.8	7.6	8.0		
(15) Precipitation	PrecAvg	37.8	2.7	2.4	3.3	3.1	3.7	3.9	3.9	3.3	3.1	2.5	2.9	2.8		
	PrecMax	57.8	6.3	7.0	6.4	5.5	6.6	10.3	8.7	7.9	10.1	4.8	11.1	8.5		
	PrecMin	25.8	0.8	0.6	0.9	0.5	1.3	0.6	0.7	1.0	0.3	0.2	0.4	0.5		
	PrecStd	6.2	1.2	1.2	1.3	1.2	1.5	1.9	1.7	1.7	1.8	1.1	1.6	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	66.8	75.8	82.7	86.7	90.5	92.1	91.6	89.1	81.4	71.3	64.3		
		MCWB	55.9	54.5	58.8	62.3	69.1	72.2	74.0	73.0	70.0	67.5	59.2	56.7		
	2%	DB	58.1	60.3	69.9	77.5	83.4	87.0	88.7	88.0	85.0	76.2	66.2	59.1		
		MCWB	53.4	51.9	55.8	59.2	66.6	71.2	72.7	71.7	69.5	63.9	55.6	53.4		
	5%	DB	52.8	54.3	64.0	73.2	80.3	84.5	86.1	85.0	81.2	72.4	62.3	54.3		
		MCWB	48.5	47.1	52.5	57.9	65.5	70.0	71.3	70.5	67.9	61.8	53.8	49.6		
	10%	DB	46.3	48.3	58.1	68.6	76.6	81.7	83.4	82.3	77.8	68.4	58.3	49.4		
		MCWB	41.9	42.3	49.7	55.2	63.8	68.7	69.9	69.1	66.3	60.3	51.6	45.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	57.9	60.8	66.1	71.8	75.4	77.4	76.4	74.6	70.5	62.3	58.7		
		MCDB	62.2	64.8	72.2	77.5	83.0	86.0	87.4	85.9	83.6	77.7	67.5	62.6		
	2%	WB	54.3	53.3	57.8	62.5	69.4	73.6	75.3	74.5	72.1	67.3	58.4	54.3		
		MCDB	57.8	58.3	65.8	72.2	79.3	83.1	84.3	83.0	80.0	72.8	64.0	57.4		
	5%	WB	49.0	47.7	54.7	60.2	67.6	72.1	73.7	73.0	70.2	64.3	55.2	50.3		
		MCDB	52.3	53.7	62.3	69.4	76.8	81.2	81.8	80.7	77.2	69.9	60.7	53.6		
	10%	WB	42.2	42.7	50.4	57.8	65.6	70.6	72.3	71.4	68.5	61.1	51.9	45.0		
		MCDB	45.7	48.0	57.4	65.9	74.0	78.8	79.9	78.8	75.0	67.7	57.4	49.0		
(35) Mean Daily Temperature Range	MDBR		14.1	15.7	17.9	20.6	19.8	18.4	18.0	18.0	18.7	18.0	15.8	13.2		
	5% DB	MCDBR	19.0	24.0	25.7	26.1	23.9	21.5	20.3	20.9	22.2	23.1	22.0	19.6		
		MCWBR	15.6	18.1	15.3	13.7	11.2	9.1	8.0	8.3	9.4	12.2	14.8	15.8		
	5% WB	MCDBR	18.0	22.2	22.1	22.1	20.2	18.5	17.2	17.0	17.4	18.5	19.0	18.6		
		MCWBR	15.8	18.2	15.0	13.6	10.4	8.8	8.1	8.2	8.5	11.5	14.8	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.326	0.352	0.380	0.421	0.452	0.457	0.454	0.387	0.362	0.318	0.300		
	taud		2.461	2.374	2.380	2.331	2.227	2.177	2.179	2.175	2.339	2.412	2.514	2.512		
	Ebn at Noon		273	281	285	283	272	263	260	258	269	264	266	266		
	Edh at Noon		25	31	35	39	44	47	46	45	36	30	23	22		
(44) All-Sky Solar Radiation	RadAvg		477	750	1051	1430	1671	1812	1826	1653	1348	896	606	403		
	RadStd		44	108	101	142	141	148	126	96	134	87	82	43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	N/A
(47) Regional (40 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+106

Nomenclature: See separate page